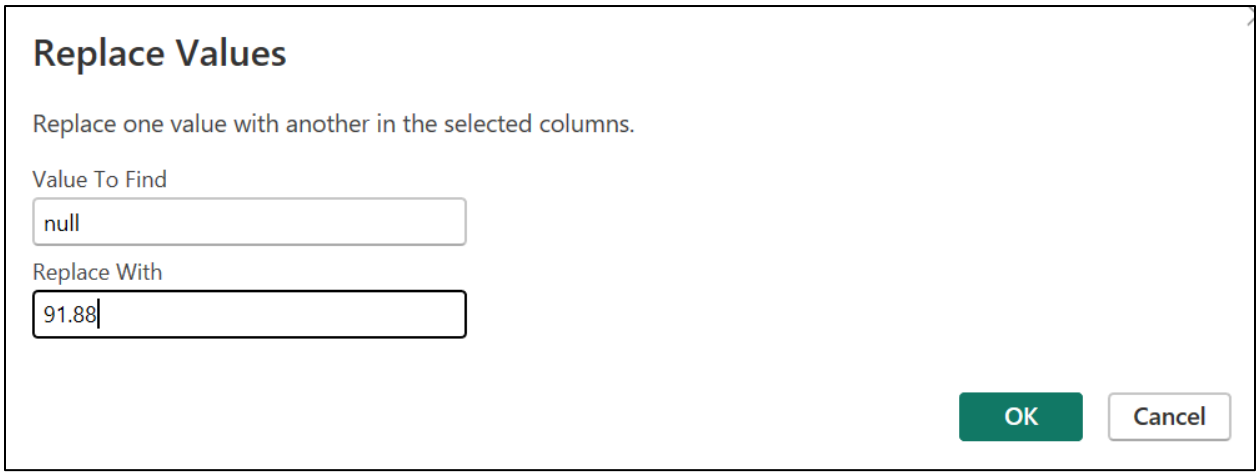
Project: Power BI Module End Project

Case Study Title: Fleet Performance & Delivery Efficiency Dashboard

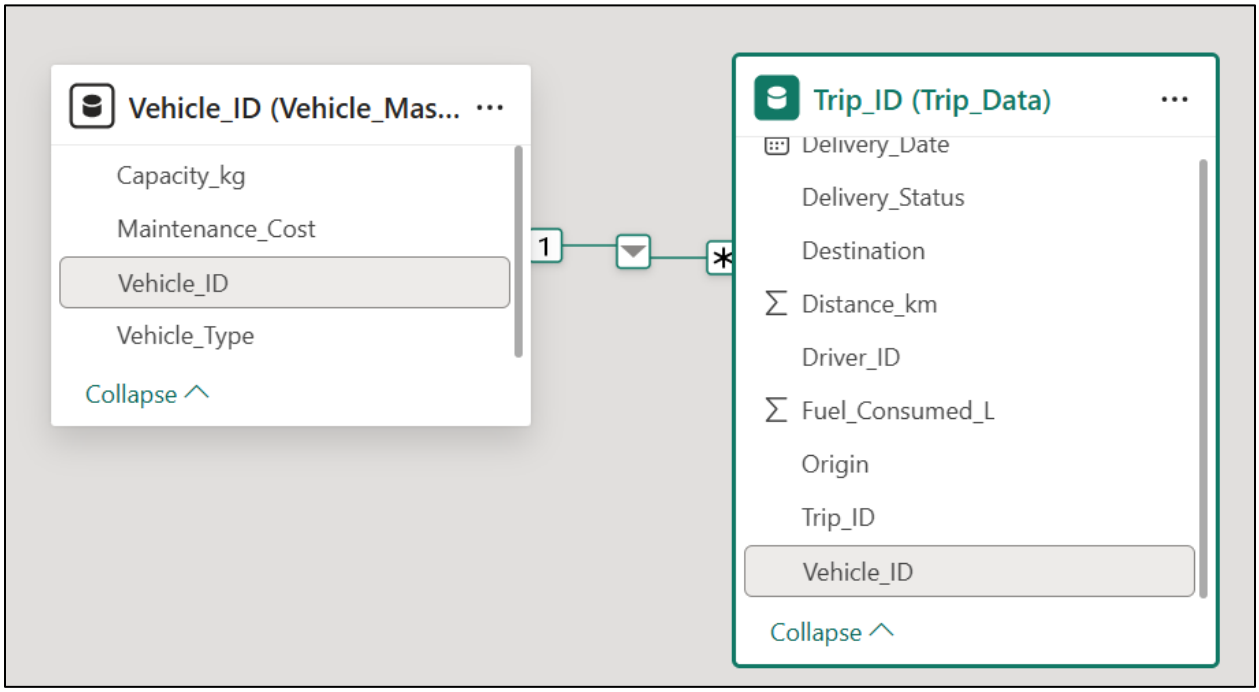
Tasks:

1. Data Cleaning & Modeling:

- Fix missing fuel consumption values (use mean imputation)



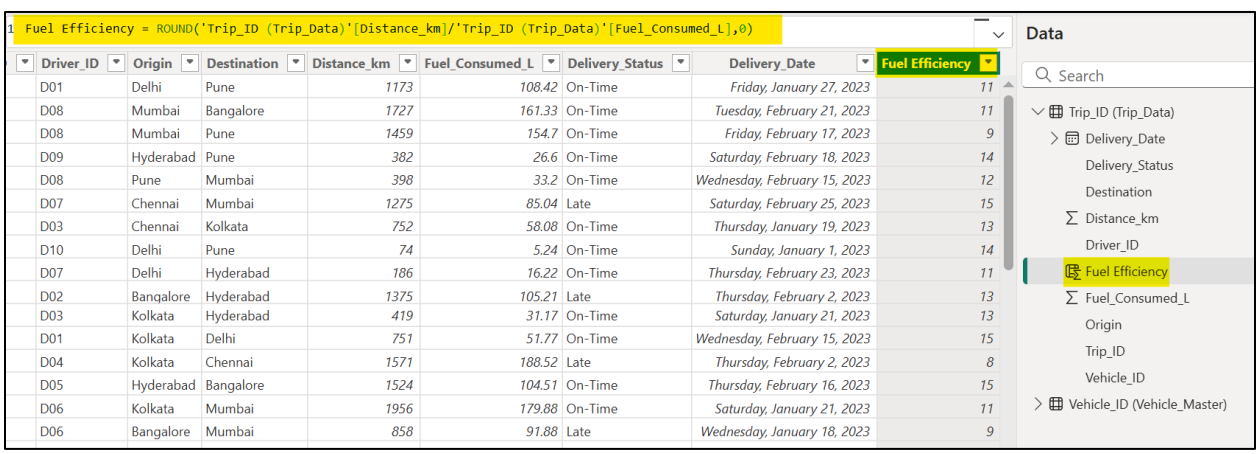
- Create a relationship using Vehicle_ID between Trip_Data and the Vehicle Master table.



2. DAX Measures:

- **Fuel Efficiency = Distance / Fuel Consumed**

```
1 Fuel Efficiency = 'Trip_ID (Trip_Data)'[Distance_km]/'Trip_ID (Trip_Data)'[Fuel_Consumed_L]
```



Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status	Delivery_Date	Fuel Efficiency
D01	Delhi	Pune	1173	108.42	On-Time	Friday, January 27, 2023	11
D08	Mumbai	Bangalore	1727	161.33	On-Time	Tuesday, February 21, 2023	11
D08	Mumbai	Pune	1459	154.7	On-Time	Friday, February 17, 2023	9
D09	Hyderabad	Pune	382	26.6	On-Time	Saturday, February 18, 2023	14
D08	Pune	Mumbai	398	33.2	On-Time	Wednesday, February 15, 2023	12
D07	Chennai	Mumbai	1275	85.04	Late	Saturday, February 25, 2023	15
D03	Chennai	Kolkata	752	58.08	On-Time	Thursday, January 19, 2023	13
D10	Delhi	Pune	74	5.24	On-Time	Sunday, January 1, 2023	14
D07	Delhi	Hyderabad	186	16.22	On-Time	Thursday, February 23, 2023	11
D02	Bangalore	Hyderabad	1375	105.21	Late	Thursday, February 2, 2023	13
D03	Kolkata	Hyderabad	419	31.17	On-Time	Saturday, January 21, 2023	13
D01	Kolkata	Delhi	751	51.77	On-Time	Wednesday, February 15, 2023	15
D04	Kolkata	Chennai	1571	188.52	Late	Thursday, February 2, 2023	8
D05	Hyderabad	Bangalore	1524	104.51	On-Time	Thursday, February 16, 2023	15
D06	Kolkata	Mumbai	1956	179.88	On-Time	Saturday, January 21, 2023	11
D06	Bangalore	Mumbai	858	91.88	Late	Wednesday, January 18, 2023	9

- **On-Time Delivery % = On-Time Trips / Total Trips:**

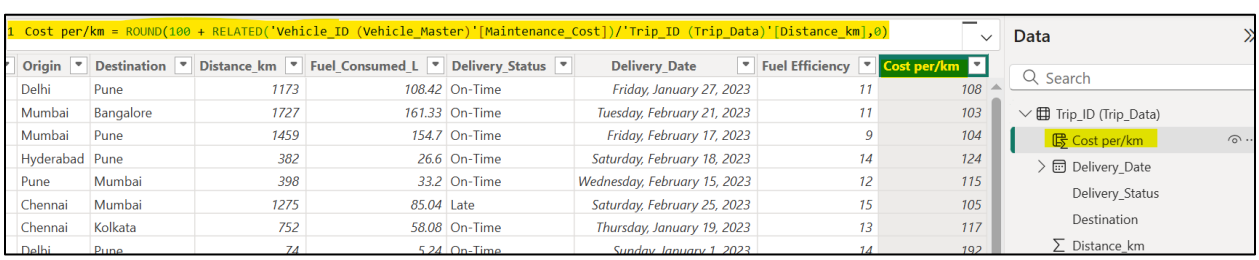
  To do this, calculate the number of "On-Time" entries in the table and the total entries in the Trip_Data table, and then divide both results.

```
On-Time Delivery = CALCULATE(COUNT('Trip_ID (Trip_Data)'[Delivery_Status]),'Trip_ID (Trip_Data)'
[Delivery_Status]="On-Time")/COUNTROWS('Trip_ID (Trip_Data)')
```

- **Cost per km = (fuel cost + Maintenance Cost) / Distance**

Note fuel cost =100

```
Cost per/km = ROUND(100 + RELATED('Vehicle_ID (Vehicle_Master)'[Maintenance_Cost])/'Trip_ID (Trip_Data)'[Distance_km],0)
```

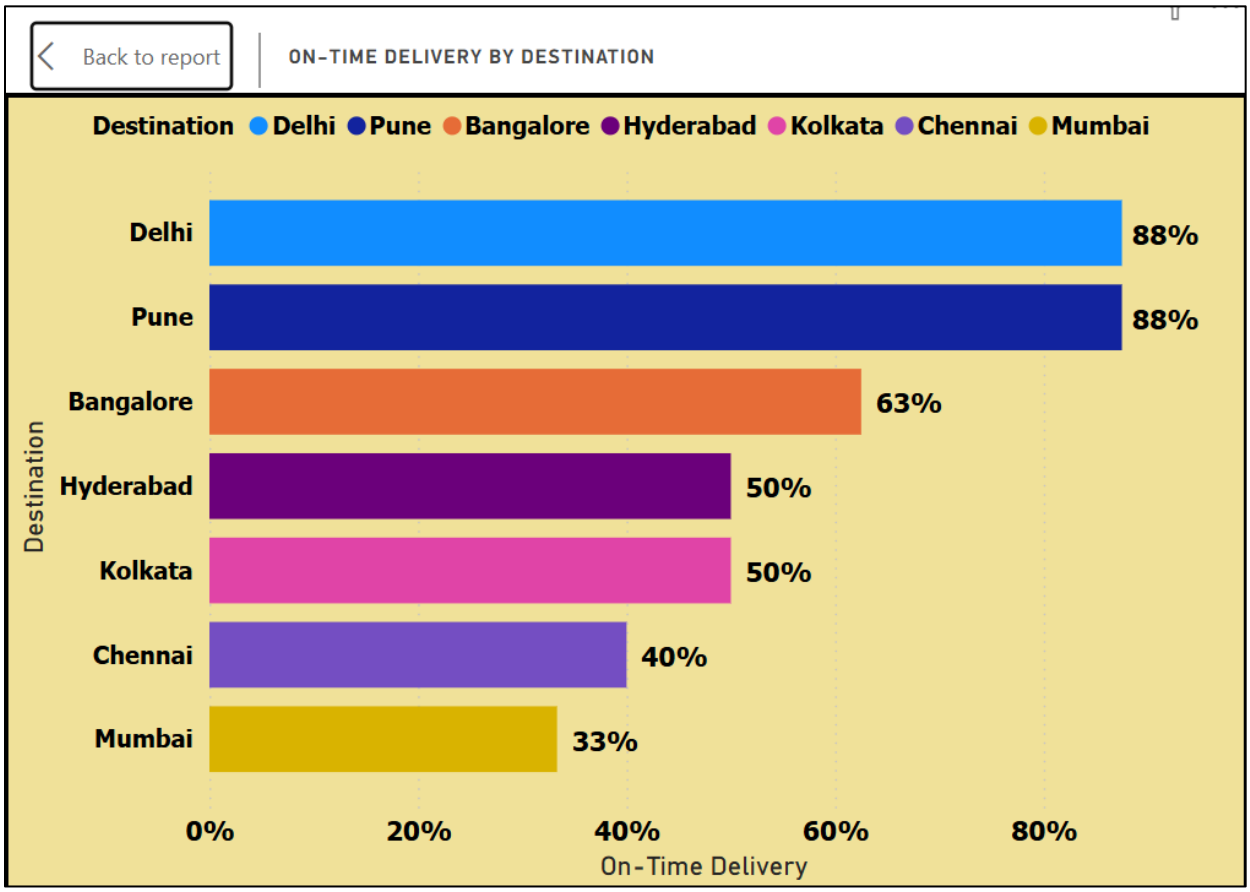


```
Cost per/km = ROUND(100 + RELATED('Vehicle_ID (Vehicle_Master)'[Maintenance_Cost])/'Trip_ID (Trip_Data)'[Distance_km],0)
```

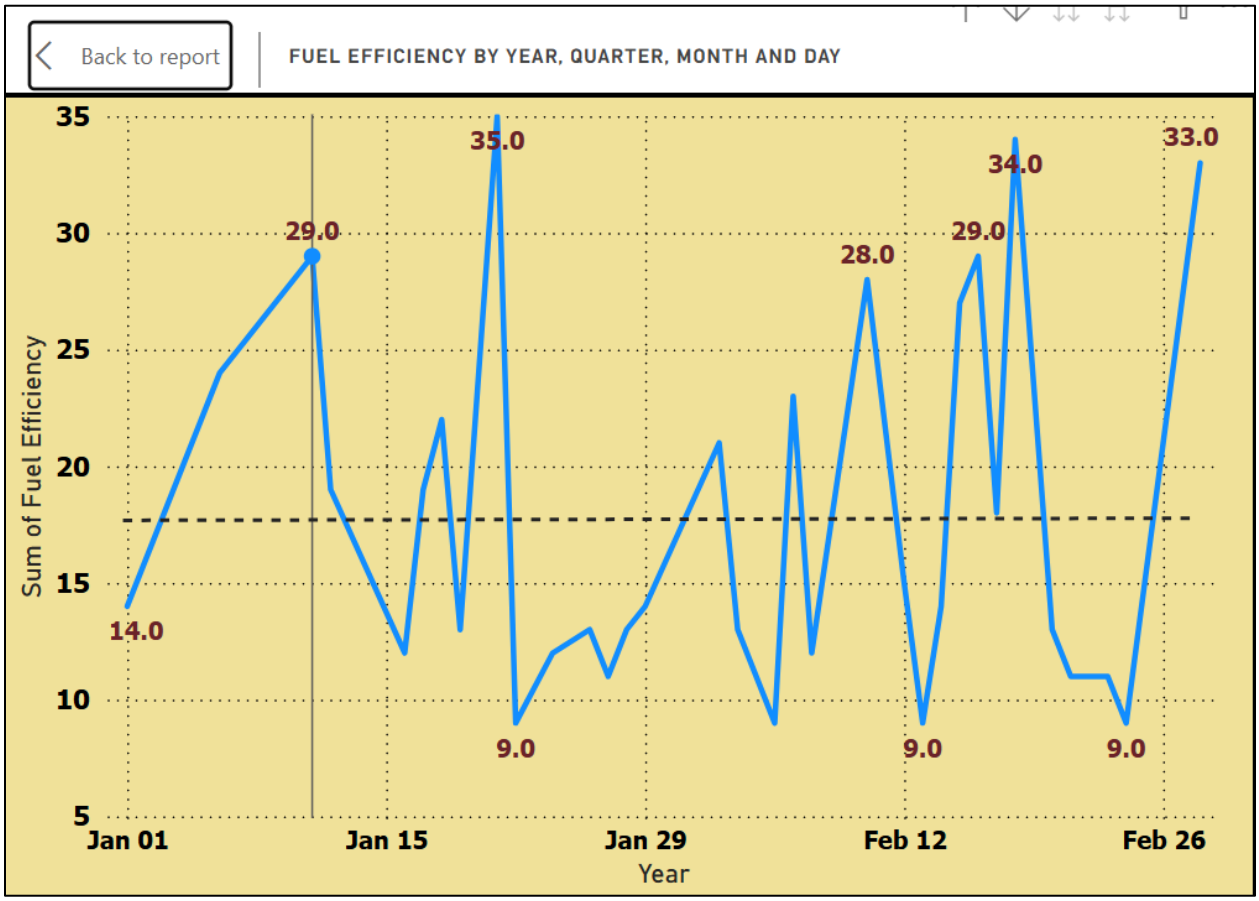
Origin	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status	Delivery_Date	Fuel Efficiency	Cost per/km
Delhi	Pune	1173	108.42	On-Time	Friday, January 27, 2023	11	108
Mumbai	Bangalore	1727	161.33	On-Time	Tuesday, February 21, 2023	11	103
Mumbai	Pune	1459	154.7	On-Time	Friday, February 17, 2023	9	104
Hyderabad	Pune	382	26.6	On-Time	Saturday, February 18, 2023	14	124
Pune	Mumbai	398	33.2	On-Time	Wednesday, February 15, 2023	12	115
Chennai	Mumbai	1275	85.04	Late	Saturday, February 25, 2023	15	105
Chennai	Kolkata	752	58.08	On-Time	Thursday, January 19, 2023	13	117
Delhi	Pune	74	5.24	On-Time	Sunday, January 1, 2023	14	192

3. Visualization:

- **Bar chart:** On-Time Delivery % by Destination.



- **Line chart:** Fuel efficiency trend by delivery date.



- **Cards visualization:**

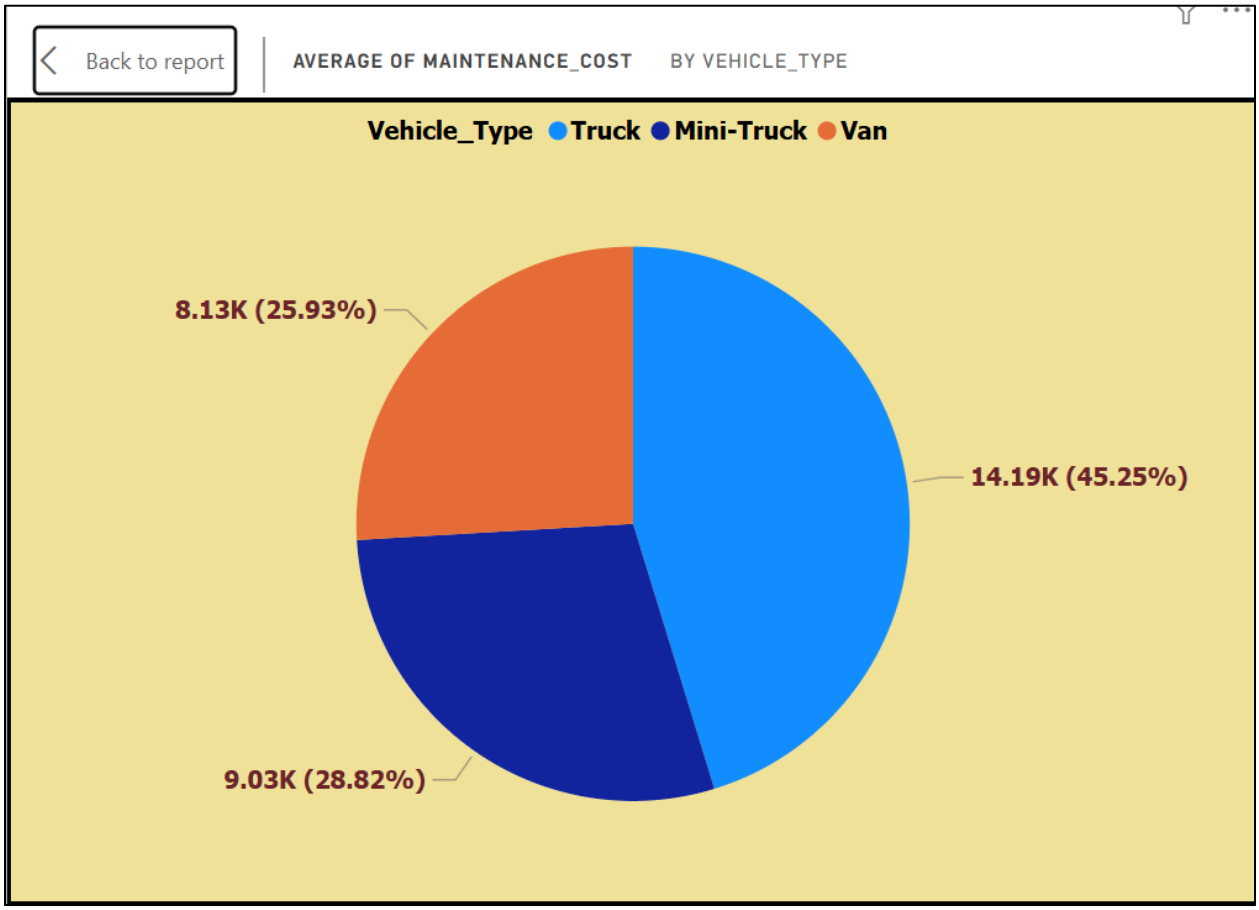
  - **Avg. Delivery Time**
    Note: *I try my best as mentioned below, literally I don't know how to find the correct method.*

```
Avg. Delivery Time = AVERAGE('Trip_ID (Trip_Data)'[Delivery_Date])
```

- **Average cost per km**

```
Average cost per km = AVERAGE('Trip_ID (Trip_Data)'[Cost per/km])
```

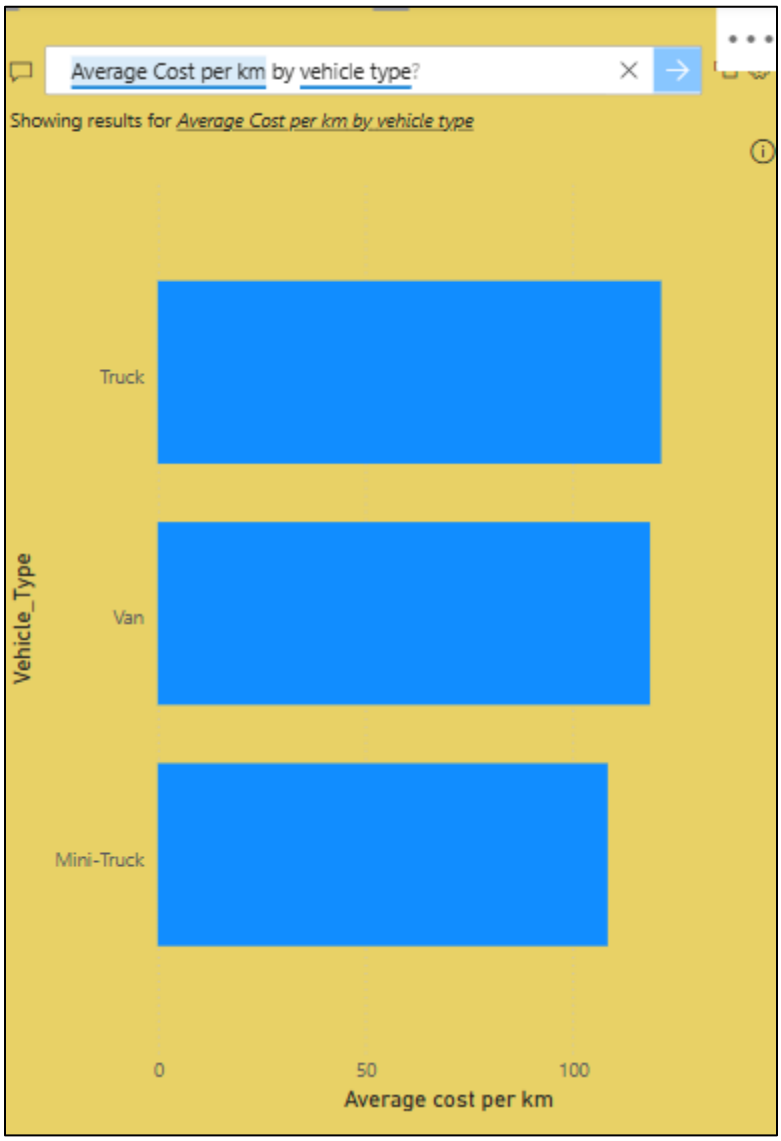
- **Pie chart:** Vehicle type vs Average maintenance cost



4. AI-Powered Visuals

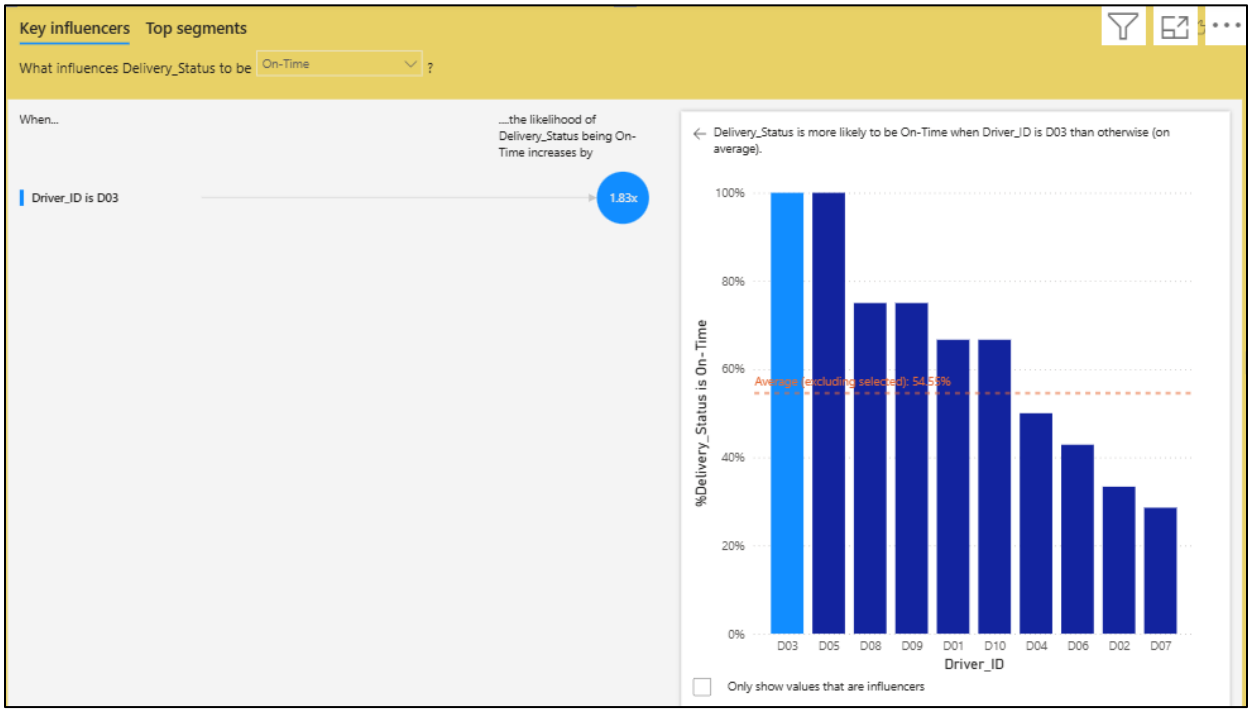
❖ **Q&A Visual:**

➢ Add Q&A visual → Prompt: "Average Cost per km by vehicle type?"



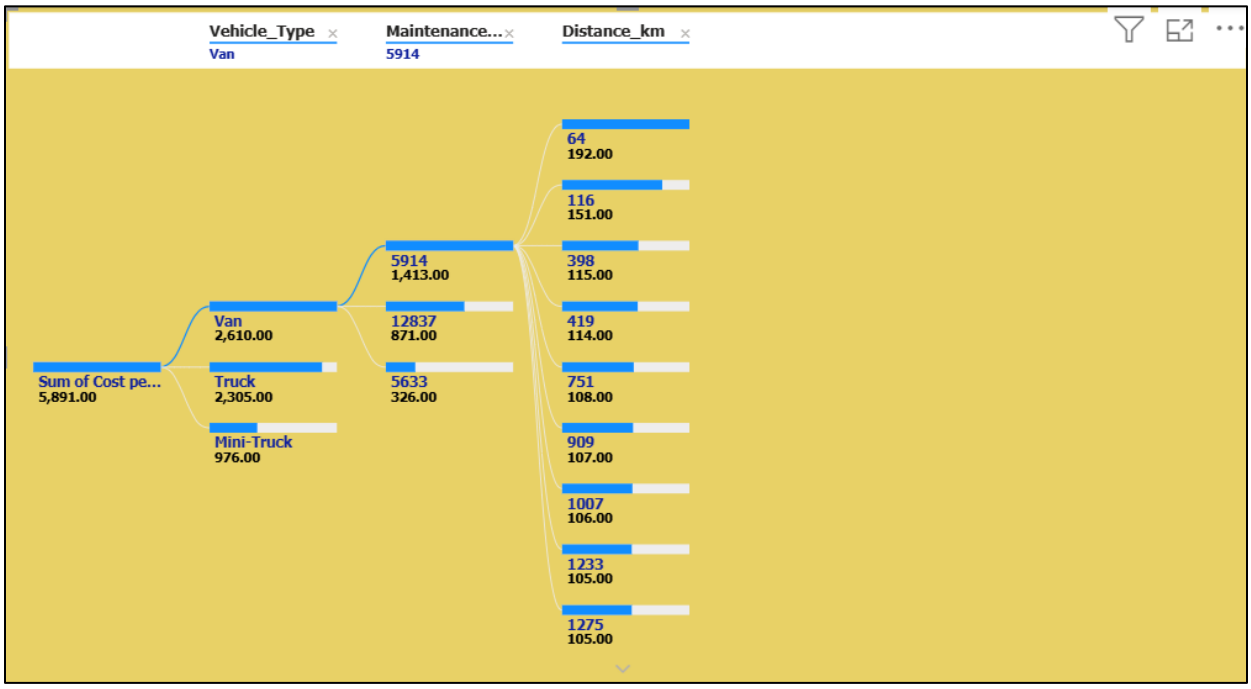
❖ Key Influencers Visual:

➢ Add Delivery Status in the Analyze and explain by -distance in km,vehicle_type, Driver_ID

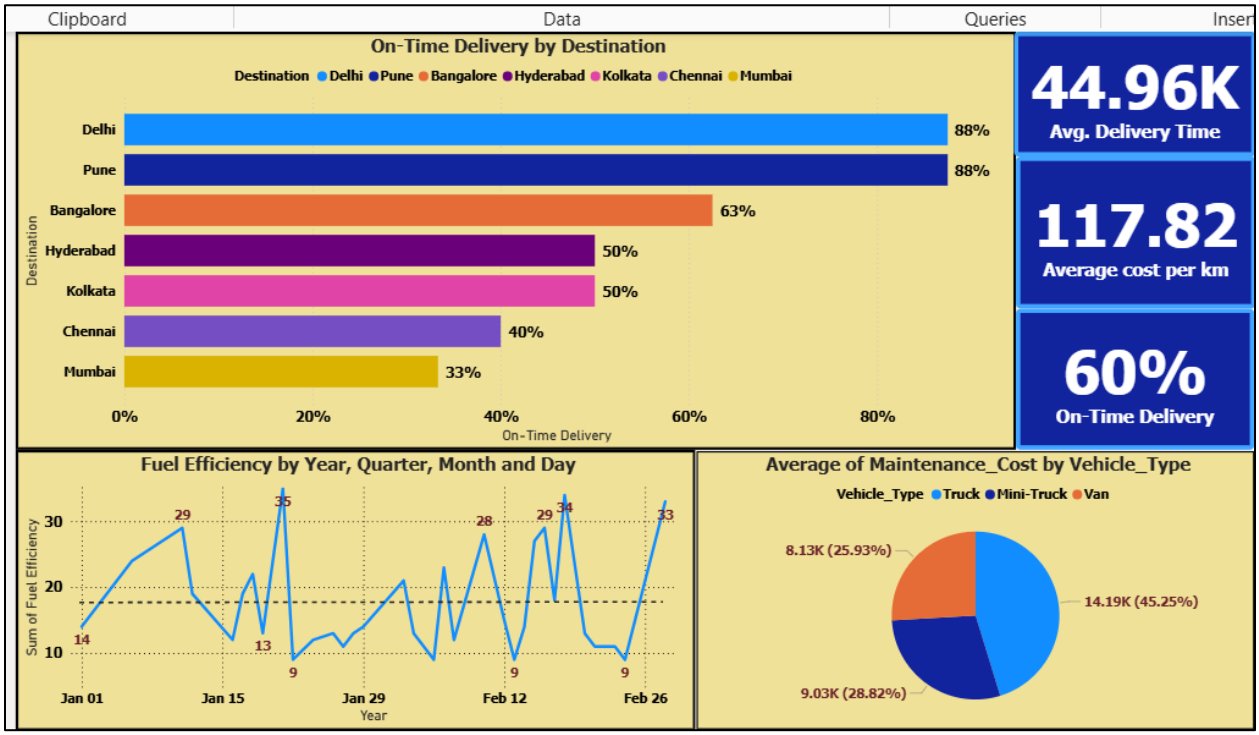


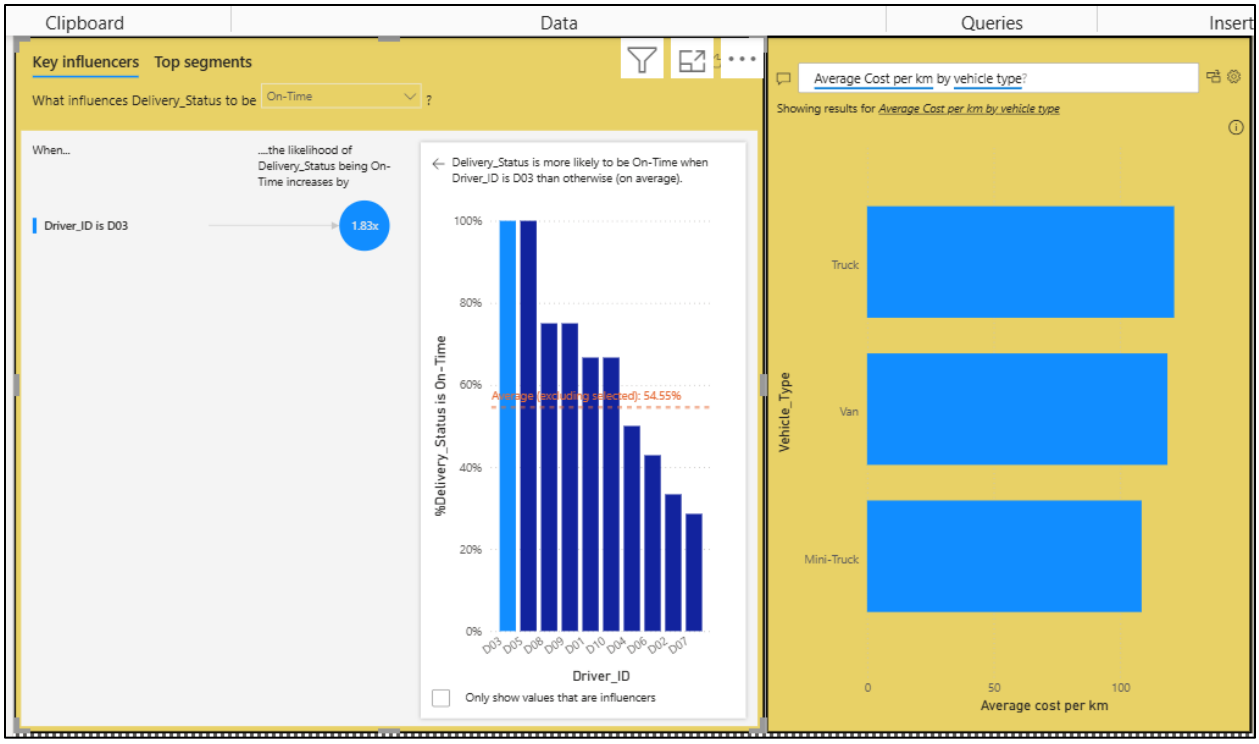
❖ **Decomposition Tree (AI Visual):**

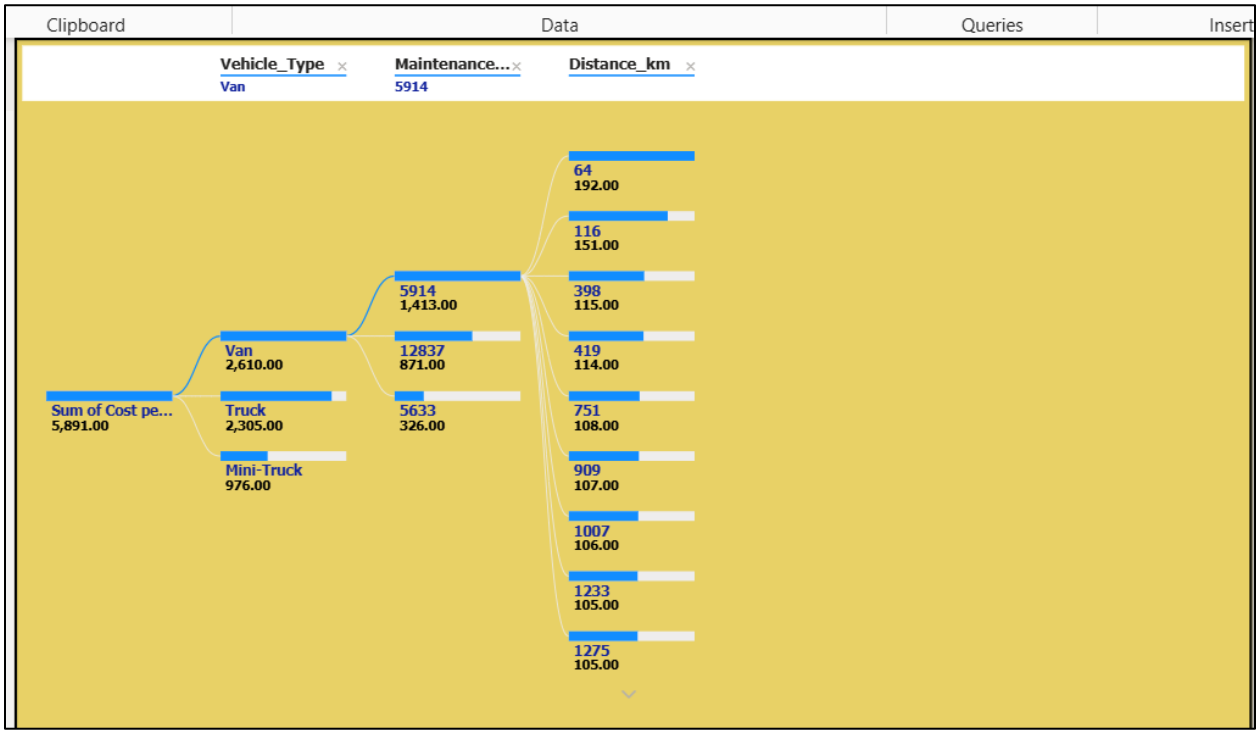
➢ Analyze Cost per km → explained by Vehicle Type, Maintenance_cost and Distance_km.



Dashboard




Clipboard
Data
Queries
Insert
Key influencers
Top segments
What influences Delivery_Status to be On-Time ?
When...
...the likelihood of Delivery_Status being On-Time increases by
Driver_ID is D03
1.83x
Delivery_Status is more likely to be On-Time when Driver_ID is D03 than otherwise (on average).
100%
80%
60%
40%
20%
0%
%Delivery_Status is On-Time
Average (excluding selected): 54.55%
D03 D05 D08 D09 D01 D10 D04 D06 D02 D07
Driver_ID
Only show values that are influencers
Average Cost per km by vehicle type?
Showing results for Average Cost per km by vehicle type
Truck
Van
Mini-Truck
Vehicle_Type
0
50
100
Average cost per km


Clipboard
Data
Queries
Insert
Vehicle_Type
Van
Maintenance...
5914
Distance_km
Sum of Cost pe...
5,891.00
Van
2,610.00
Truck
2,305.00
Mini-Truck
976.00
5914
1,413.00
12837
871.00
5633
326.00
64
192.00
116
151.00
398
115.00
419
114.00
751
108.00
909
107.00
1007
106.00
1233
105.00
1275
105.00